

Nexum Suum – Boiler Troubleshooting SOP

■ Page1: Pressure, Water Level, Combustion, and Leak Troubleshooting Tree

If Pressure Too High or Relief Valve Lifting:

- Check Pressuretrol or High-Limit Control Setpoint.
- Verify Operating vs Manual Reset Controls – Miswiring common.
- Ensure Modulation or Burner Control Operating Correctly.

If Low Water or LWCO Trip:

- Inspect Feedwater Pump and Make-up Valve.
- Check LWCO Probe for Fouling or Electrical Failure.
- Verify Proper Blowdown Schedule – Sediment can insulate probes.

If Combustion Fluctuates or Poor Flame:

- Test Flame Sensor and Igniter Condition.
- Check Fuel Supply Pressure and Regulator Function.
- Inspect Burner Head for Carbon Build-up or Corrosion.

If Leaks Found Around Boiler:

- Inspect Manway Gasket, Sight Glass, Blowdown Valve.
- Look for Signs of Tube Sheet or Bundle Failure.
- Check Condensate Tank and Surge Vessel Overflow.

Nexum Suum – Boiler Troubleshooting SOP (Cont'd)

■ Page2: Symptom-to-Action Mapping + Real-World Thresholds

Symptom	Action
Pressure Exceeding Limit	Check pressuretrol, high limit control, modulating burner.
LWCO or Low Water Alarm	Verify probe condition, blowdown history, pump runtime.
Unstable Flame or Shutdown	Inspect igniter, sensor, gas pressure regulator.
Leak Around Boiler Gasket	Tighten or replace gasket, inspect flange surfaces.
Surge or Condensate Overflow	Tank undersized or vented incorrectly, check return trap.

■ Operating Thresholds:

- Steam Pressure: 10–150 psi (varies by system)
- Feedwater Temp: 180–200°F for high efficiency
- Blowdown Frequency: 1x per shift or per schedule
- Stack Temp: 300–500°F depending on load
- LWCO Cutoff: Triggered if <1.5" water column (probe) or float drop
- Condensate Return Rate: Match boiler GPM (±10%)
- Make-up Water Conductivity: <50 µS/cm recommended pH: 10.5–11.5 | Hardness: <1 ppm | TDS: <3,500 ppm